

AI Trader: SOTA Quant Implementation Plan

Target Root: D:\Ai trader | Architecture: Low-Latency Dual-Loop Market Maker (2026)

1. Executive Tooling Scorecard & Upgrade Matrix

Subsystem	Previous	SOTA Upgrade	Score	Rationale
Risk Model	Gaussian VaR	Cornish-Fisher VaR	5.0 → 9.0	Corrects 50-300% tail risk underestimation using empirical skewness & kurtosis.
Indian Broker	Finvasia Shoonya	DhanHQ v2 API	5.5 → 9.0	Sub-15ms execution, institutional websocket stability, zero morning auth drops.
Flow Threshold	Static Z-Score	Rolling 98th %ile	4.0 → 8.5	Adapts to volatility regimes; stops false halts during quiet market hours.
Regime Model	HMM (Markov)	BOCPD	6.5 → 8.5	Bayesian changepoints detect shifts in 1-3 ticks; no label switching.
Ring Buffer	Standard NumPy	Mirrored 2N Buffer	7.0 → 9.0	Zero-copy contiguous window slicing; O(1) Welford scalar variance math.
Dashboard	Jinja2 + Canvas	TradingView Charts	6.0 → 9.0	45KB Lightweight Charts for OHLCV; keep Canvas for L2 DOM ladder.
Tick Storage	SQLite Only	DuckDB + Parquet	7.5 → 9.5	SQLite for atomic live state; columnar Parquet for 50x backtest speed.

Phase 0: Security & Codebase Integrity

- Task 0.1: Port Rebinding: Rebind dashboard from port 8000 to 8002 in cli.py to prevent port collisions.
- Task 0.2: CORS Whitelisting: Allow localhost:8002 in dashboard/app.py.
- Task 0.3: Scrub Mock Data: Delete fake Math.sin synthetic confidence bars from index.html.

Verification Gate 0: Dashboard boots clean on port 8002 with live WebSocket feed only.

Phase 1: Risk Engine Overhaul (Cornish-Fisher VaR)

- Task 1.1: Cornish-Fisher VaR: Implement CF expansion in risk_manager.py using empirical skew & excess kurtosis.
- Task 1.2: Adaptive Flow Gating: Calculate rolling 98th percentile for toxic order flow imbalance (OFI).
- Task 1.3: Max Drawdown Circuit Breaker: Enforce 2% intraday equity peak halt via data/STOP flag.

Verification Gate 1: pytest tests/test_risk_manager.py confirms non-NaN fat-tailed VaR scaling.

Phase 2: Broker Migration (DhanHQ v2 API)

- Task 2.1: DhanHQ Adapter: Implement dhanhq_adapter.py for sub-15ms NSE F&O; order execution.
- Task 2.2: CCXT Pro Binance Stream: Verify tri-stream websocket endpoint (wss://stream.binance.com:9443/ws).

Verification Gate 2: Dry-run order placed on DhanHQ with verified response time under 15ms.

Phase 3: High-Frequency Data Structures (2N Buffer & BOCPD)

- Task 3.1: Mirrored 2N Buffer: Implement contiguous zero-copy double buffer in live_runner.py.
- Task 3.2: BOCPD Regime Detection: Replace HMM with Bayesian Online Changepoint Detection (1-3 tick response).
- Task 3.3: Multi-Level OFI: Weight top 5 depth levels with exponential decay factor in market_maker.py.

Verification Gate 3: 1M ticks processed with zero garbage collection spikes and < 5MB memory growth.

Phase 4: Tick Storage Hybridization (DuckDB + Parquet)

- **Task 4.1: Hourly Parquet Flush:** Write completed tick buffers to data/ticks/{date}.parquet.
- **Task 4.2: DuckDB Analytical Engine:** Query parquet files in-process with 50x throughput over SQLite.

Verification Gate 4: 1-month historical backtest query executes in < 100ms via DuckDB.

Phase 5: Dashboard Visualization Upgrade

- **Task 5.1: TradingView Charts:** Embed Lightweight Charts v4.2 (45KB) for responsive OHLCV rendering.
- **Task 5.2: Retain Canvas for L2 DOM:** Keep pure HTML5 canvas for sub-millisecond orderbook ladder updates.

Verification Gate 5: 60 FPS locked dashboard with zero frame drops during heavy order flow.